NOTE **The mathematical convenience of entropy**

This is a mathematically elegant procedure that can be applied to many situations of decision-making under uncertainty. We can think of this entropy-based search procedure as a kind of the search for the truth, in which we aim to take the action that takes us as close to the truth as possible, by maximally reducing uncertainty.

6.1.3 Binary search using entropy

To understand entropy-based search further, we will now see how this procedure manifests itself in one of the classic algorithms in computer science, binary search. You are most likely already familiar with this algorithm, so we won't be going into much detail here. For an excellent and beginner-friendly explanation of binary search, I recommend Chapter 1 of Aditya Bhargava's *Grokking Algorithms* (Manning, 2016).

On a high level, we employ binary search when we want to look for the position of a specific targeted number within a sorted list.

IMPORTANT **How binary search works**

The idea behind binary search is to look at the middle element of the list and compare it to the target. If the target is smaller than the middle element, then we know to only look at the first half of the list; otherwise, we look at the second half. We repeat this process of halving the list until we find the target.

Consider a concrete example in which we have a sorted list of 100 elements $[x_1, x_2, \ldots, x_{100}]$, and we'd like to find the location of a given target z, assuming that z is indeed inside the sorted list.

As illustrated by Figure 6.5, binary search works by divided the list into two halves: the first 50 elements and the last 50 elements. Since we know that this list is sorted:

- If our target z is less than the 50th element x_{50}, then we only need to consider the first 50 elements, since the last 50 elements are all greater than target z, and similarly
- If our target is greater than x_{50}, then we only need to look at the second half of the list.

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